# PLACEPRO — STUDENT MANAGEMENT & PLACEMENT INTELLIGENCE PORTAL

Milestone 1 Deliverable: Software Requirements Specification (SRS). Prepared strictly in accordance with Milestone 1 (Requirement Phase) of the Project Milestone Guide for Students—defining what the system will and will not do before system design or coding begins.

| Document Control Field | Details |
|---|---|
| Project Name | PlacePro — Student Management & Placement Intelligence Portal |
| Milestone Phase | Milestone 1 — Requirement (What are we building and for whom?) |
| Group No. | Group 171 |
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## 1. Problem Statement

Technical trainers and placement teams conduct frequent readiness exams and mock interviews using spreadsheets that lack a fixed or uniform column structure, making it slow and error-prone to consolidate student marks, preserve qualitative feedback, and track historical progress. Currently, there is no unified portal that can ingest arbitrary trainer spreadsheets without losing custom fields, provide a 360-degree view of every student alongside batch-level analytics, and dynamically filter candidates based on company-specific skill requirements (such as Infosys DSA criteria or Cognizant Java and Spring Boot criteria).

## 2. Introduction & Purpose of the SRS

### 2.1 Purpose

This Software Requirements Specification (SRS) formally defines what the PlacePro Student Management & Placement Intelligence Portal must do and how it must behave. It serves as the foundational agreement between the five-member student development team and the project mentor/trainer prior to system design (Milestone 2) and software development (Milestone 3).

### 2.2 Product Overview

PlacePro is a role-based placement intelligence platform with three dedicated portals (Technical Trainer Portal, Student Portal, and Admin Portal). It enables technical trainers to upload unstructured exam and mock interview spreadsheets, retains the original upload logs alongside normalized skill scores, generates a real-time 360-degree performance profile for each student, and allows placement staff to filter and shortlist students against company-specific skill requirements.

### 2.3 Key Domain Terminology

| Term | Definition in PlacePro |
|---|---|
| Unstructured Assessment Sheet | An Excel/CSV file uploaded by a technical trainer (for a daily/weekly exam or mock interview) whose column headers, max marks, and extra remarks columns may vary from sheet to sheet. |
| Column Mapping & Preview | A review step during file upload where the trainer associates spreadsheet headers with student identity, known skills, or qualitative review sections, while preserving any extra columns. |
| Original Upload Log | The unaltered record of an uploaded exam or mock interview spreadsheet preserved so trainers can analyze individual assessment events at any time. |
| Student 360° View | A unified scorecard for an individual student showing subject-wise progress, domain readiness (DSA, Core CS, Development, Communication), strong/weak areas, and historical feedback. |
| Placement Readiness (PR / NPR) | A classification indicator showing whether a student is currently Placement Ready (PR) or Not Placement Ready (NPR) based on aggregated evaluations. |
| Dynamic Skill / Column Parameter | A skill or evaluation metric (e.g., DSA, DBMS, OS, CN, OOPS, Java, |

| | |
|---|---|
| | Spring Boot, Communication) that an Admin can add or configure so trainers can score and filter students on it. |

## 3. User Roles & Portal Responsibilities

The system supports three distinct user roles, each with a dedicated portal and strict access boundaries:

| User Role | Portal | Core Responsibilities & Access Scope |
|---|---|---|
| 1. Technical Trainer (and TPO / Placement Staff) | Technical Trainer Portal | • View batch-wide matrix, placement readiness statistics (PR vs. NPR), and top/focus students.<br>• Upload unstructured Exam Excel/CSV sheets and Mock Interview Excel/CSV sheets.<br>• Map spreadsheet columns during upload while preserving the original spreadsheet log.<br>• Analyze individual exams/interviews as well as drill down into any student's complete 360° profile.<br>• Filter and sort students dynamically based on company-specific skill requirements (e.g., Infosys strong DSA; Cognizant Java + Spring Boot). |
| 2. Student | Student Portal | • View only their own personal and academic details (Name, Email, Roll Number, Batch, CGPA).<br>• Monitor their own marks and performance in individual exams and mock interviews.<br>• View visual progress graphs across subjects and skill domains.<br>• Read qualitative trainer feedback (Strong Areas, Weak Areas/Areas to Target, Behavior Review, and improvement guidance). |
| 3. Administrator (Admin) | Admin Portal | • Manage user accounts, role assignments (Admin, Trainer, Student), and student batch records.<br>• Dynamically add and edit evaluation parameters/columns (skills, domains, and custom attributes) so the system adapts to new training topics or company requirements.<br>• Configure company profiles, skill weightages, and minimum eligibility cutoffs.<br>• View system audit logs and manage placement drive records. |

## 4. Core Use Cases (Actor -> Goal -> Steps)

Below are the 10 core use cases describing how each actor interacts with PlacePro to achieve their goals:

- **UC-01: Role-Based Sign-In —** Actor: User (Trainer, Student, or Admin) -> Goal: Log into the portal and access role-specific dashboard features. -> Steps: Open login page -> Enter institutional email and password -> Click 'Login' -> System verifies credentials and role -> Redirect user to Trainer Portal, Student Portal, or Admin Portal.
- **UC-02: Upload & Map an Unstructured Exam Sheet —** Actor: Technical Trainer -> Goal: Upload a daily/weekly exam Excel sheet with arbitrary columns and integrate scores into the main system while keeping the original log. -> Steps: Open Trainer Portal -> Select 'Upload Exam Sheet' and choose Excel/CSV file -> Preview detected column headers and sample rows -> Map Roll Number column, map score columns to skills with maximum marks, and retain any extra/unrecognized columns -> Confirm import -> System saves original exam log and updates student skill scores.
- **UC-03: Upload & Map a Mock Interview Sheet —** Actor: Technical Trainer -> Goal: Upload a mock interview spreadsheet containing technical scores (DSA, Core CS, Communication) and qualitative reviews (comments, behavior review). -> Steps: Open Trainer Portal -> Select 'Upload Mock Interview Sheet' and choose file -> Map Roll Number, technical rating columns, strengths/weaknesses comments, and behavior review column -> Confirm import -> System logs the original interview sheet and updates each student's 360° profile.

- **UC-04: View Batch Matrix & Pinpoint Focus Students —** Actor: Technical Trainer -> Goal: Monitor overall batch readiness and identify students who need targeted training intervention. -> Steps: Open Trainer Dashboard -> View batch readiness breakdown (Placement Ready 'PR' vs. Not Placement Ready 'NPR' counts and percentages) -> Inspect the leaderboard and flagged focus students -> Select a focus student to view their weak areas.
- **UC-05: Inspect Complete Student 360° View —** Actor: Technical Trainer -> Goal: Analyze a single student's holistic performance across all historical exams and mock interviews. -> Steps: Search or click on a student (e.g., from the Trainer Dashboard list) -> Open the 360° Student View -> Inspect subject progress graphs (DSA, OS, CN, DBMS, OOPS, Communication), persona tags, strong fields, weak fields, and individual exam/interview logs.
- **UC-06: Analyze Individual Exam or Mock Interview Log —** Actor: Technical Trainer -> Goal: Review the exact original data and remarks from a specific past exam or mock interview. -> Steps: Navigate to Assessment / Import History in the Trainer Portal -> Select a specific exam or mock interview batch -> View the student-wise scores, qualitative comments, and any custom columns that were present in that original Excel sheet.
- **UC-07: Filter & Shortlist Students for a Company Drive —** Actor: Technical Trainer / TPO -> Goal: Filter and rank students based on a visiting company's specific technical requirements. -> Steps: Open the Company Filter / Matching view -> Select a company profile or specify skill criteria (e.g., Infosys requiring high DSA problem-solving, or Cognizant requiring Java and Spring Boot) -> Apply filter -> View ranked list of eligible students and export the shortlist.
- **UC-08: Monitor Personal Progress & Trainer Feedback —** Actor: Student -> Goal: Check personal exam marks, skill graphs, and actionable improvement feedback. -> Steps: Log into Student Portal -> View personal details and overall readiness status -> Inspect recent exam performance graphs and subject breakdown -> Read Strong Areas, Weak Areas to Target, and mock interview behavior/technical remarks.
- **UC-09: Dynamically Add New Skill Parameters / Columns —** Actor: Admin -> Goal: Add a new skill or evaluation parameter to the system when a mentor or company introduces a new requirement. -> Steps: Log into Admin Portal -> Open Skill & Parameter Configuration -> Enter new parameter/skill name (e.g., 'Spring Boot' or 'System Design') and assign its domain category -> Save -> Parameter becomes immediately available for Trainer Excel mapping and Company Filtering.
- **UC-10: Configure Company Eligibility & Skill Weightages —** Actor: Admin -> Goal: Define a recruiting company's required skills, weightages, and minimum cutoffs. -> Steps: Open Admin Portal -> Navigate to Company Requirements -> Create or edit a company role (e.g., Infosys DSE or Cognizant GenC Next) -> Set minimum CGPA, required skills, percentage weightages, and minimum score cutoffs -> Save configuration.

### 4.1 Textual Use Case Diagram Summary

| Actor | Connected Use Cases in System Boundary |
|---|---|
| Technical Trainer / TPO | (UC-01) Sign-In, (UC-02) Upload Exam Excel Sheet, (UC-03) Upload Mock Interview Sheet, (UC-04) View Batch Matrix & Focus Students, (UC-05) Inspect Student 360° View, (UC-06) Analyze Individual Exam/Interview Log, (UC-07) Filter Students by Company Requirements |
| Student | (UC-01) Sign-In, (UC-08) View Personal Marks, Progress Graphs, Strong/Weak Areas & Interview Feedback |
| Admin | (UC-01) Sign-In, (UC-09) Dynamically Add/Edit Skill Parameters & Columns, (UC-10) Configure Company Criteria & Manage Users/Batches |

## 5. Functional Requirements (Per Module)

All functional requirements are organized by system module and stated using formal 'The system shall...' specifications:

### Module 1: User Authentication & Role-Based Access Control

- **FR-1.1:** The system shall allow users to log in using their institutional email address and password.
- **FR-1.2:** The system shall authenticate credentials and route the user to the Admin Portal, Technical Trainer Portal, or Student Portal based on their assigned role.
- **FR-1.3:** The system shall restrict students to viewing only their own profile, assessment scores, and feedback records.
- **FR-1.4:** The system shall allow administrators to create, update, and deactivate user accounts and assign roles.

### Module 2: Unstructured Spreadsheet Upload & Column Mapping (Exams & Mock Interviews)

- **FR-2.1:** The system shall allow technical trainers to upload Excel (.xlsx, .xls) and CSV (.csv) spreadsheets for both written/coding exams and mock interviews.

- **FR-2.2:** The system shall parse and preview the uploaded spreadsheet's column headers and sample rows before committing data to the database.
- **FR-2.3:** The system shall allow the technical trainer to map spreadsheet columns to student identifiers (Roll Number), known skill parameters, maximum marks, and qualitative review fields (Strengths, Weaknesses, Behavior Review).
- **FR-2.4:** The system shall accept and preserve any arbitrary or unmapped columns present in an uploaded Excel sheet without rejecting the file or losing data.
- **FR-2.5:** The system shall retain the complete original log of every uploaded exam and mock interview spreadsheet so individual assessments can be analyzed independently.
- **FR-2.6:** The system shall match uploaded rows to existing students using their institutional Roll Number and flag or onboard missing/ambiguous student identities safely.

### Module 3: Admin Portal & Dynamic Parameter Configuration

- **FR-3.1:** The system shall provide an Admin Portal where administrators can dynamically add new skill parameters, evaluation categories, or custom profile attributes.
- **FR-3.2:** The system shall organize skills across core domains including Programming/DSA, Core Computer Science (DBMS, OS, CN, OOPS), Software Development (e.g., Java, Spring Boot), and Communication/Soft Skills.
- **FR-3.3:** The system shall make newly added skill parameters immediately available in the Technical Trainer's Excel upload mapper and company filter menus.
- **FR-3.4:** The system shall ensure that adding a new parameter or column does not corrupt or overwrite historical exam and interview records.

### Module 4: Dynamic Skill Aggregation & Student 360° View

- **FR-4.1:** The system shall normalize marks from different exams and mock interviews onto a standard percentage scale (0–100%) based on the maximum marks of each assessment.
- **FR-4.2:** The system shall dynamically calculate each student's current skill and domain averages across all historical exams and mock interviews.
- **FR-4.3:** The system shall generate a complete 360° Student View displaying subject-wise progress graphs, overall readiness status (Placement Ready 'PR' vs. Not Placement Ready 'NPR'), and talent persona tags.
- **FR-4.4:** The system shall automatically highlight each student's Strong Fields and Weak Fields (Areas to Target) based on their aggregated skill scores and trainer remarks.
- **FR-4.5:** The system shall maintain chronological history so trainers and students can see skill progression across multiple assessments over time.

### Module 5: Technical Trainer Dashboard & Batch Analytics

- **FR-5.1:** The system shall display a batch-level dashboard showing total students, Placement Ready (PR) vs. Not Placement Ready (NPR) distributions, and domain readiness statistics.
- **FR-5.2:** The system shall display a ranked leaderboard of students and allow trainers to pinpoint focus students who require targeted training intervention.
- **FR-5.3:** The system shall allow the trainer to click on any student in the batch list to open their detailed 360° Student View.
- **FR-5.4:** The system shall allow the trainer to view a list of all historical exam and mock interview uploads and inspect the scores and remarks of any single assessment.

### Module 6: Student Self-Service Portal

- **FR-6.1:** The system shall display the logged-in student's personal academic profile (Name, Roll Number, Email, Batch, Section, CGPA).
- **FR-6.2:** The system shall display visual performance graphs showing the student's marks in recent exams and their standing across technical and communication skills.
- **FR-6.3:** The system shall display dedicated cards for 'Weak Areas / Areas to Target' and 'Strong Areas / How to Improve Further' based on trainer evaluations.
- **FR-6.4:** The system shall allow the student to view their historical exam marks and approved mock interview feedback/behavior remarks.

### Module 7: Company Requirement Configuration & Candidate Filtering

- **FR-7.1:** The system shall allow staff to create company profiles specifying required skills, skill weightages, minimum skill cutoffs, and minimum CGPA (e.g., Infosys requiring strong DSA; Cognizant requiring Java and Spring Boot).
- **FR-7.2:** The system shall allow the technical trainer to filter and sort students based on any combination of skills, domain scores, readiness status, and company requirements.
- **FR-7.3:** The system shall compute a weighted compatibility match score for each student against a selected company's requirements.
- **FR-7.4:** The system shall clearly distinguish between students who meet all mandatory cutoffs (Eligible) and those who fall short, displaying the exact reason for any shortfall.
- **FR-7.5:** The system shall allow trainers and placement staff to export the filtered and ranked candidate shortlist to a spreadsheet.

## 6. Non-Functional Requirements

The system must satisfy the following quality attributes across all six standard non-functional categories:

| Category | ID | Non-Functional Requirement Specification |
|---|---|---|
| Performance | NFR-01 | The system shall load dashboard views and execute multi-criteria company filtering queries within 3 seconds for a batch of up to 500 students, and process spreadsheet previews within 5 seconds for files up to 10 MB. |
| Security | NFR-02 | The system shall store all user passwords in encrypted/hashed form, enforce strict server-side Role-Based Access Control (Admin, Trainer, Student), and prevent unauthorized users from accessing other students' records. |
| Usability | NFR-03 | The system shall provide a clean, responsive user interface on desktop and mobile/tablet browsers, and enable a technical trainer to map and upload an unstructured Excel sheet in under 60 seconds with clear row/column error messages. |
| Scalability | NFR-04 | The system shall support the dynamic addition of new skills, custom spreadsheet columns, multiple academic batches, and hundreds of historical exam/interview logs without requiring code rewrites or downtime. |
| Reliability & Availability | NFR-05 | The system shall process spreadsheet imports atomically (ensuring failed or interrupted uploads do not leave partial or corrupted data) and maintain 99% availability during active training and placement hours. |
| Maintainability | NFR-06 | The system shall be built using a modular, well-documented codebase with clear separation between presentation, business logic, and data storage so all five team members can develop and maintain features independently. |

## 7. Project Scope & Constraints

### 7.1 Included in Scope (What the Project Will Do)

- Three role-based portals: Technical Trainer Portal, Student Portal, and Admin Portal.
- Unstructured Excel/CSV upload with interactive column mapping for both daily/weekly exams and mock interviews.
- Preservation of original uploaded spreadsheet logs alongside integrated main database records.
- Individual exam/interview analysis as well as holistic Student 360° profiles (progress graphs, strong/weak areas, readiness status, and behavior/qualitative feedback).
- Admin interface to dynamically add new skill parameters/columns and configure company eligibility criteria.

- Dynamic candidate filtering, sorting, and weighted matching based on company-specific skill requirements (e.g., Infosys DSA filter, Cognizant Java + Spring Boot filter) and shortlist export.

### 7.2 Out of Scope (What the Project Will Not Do)

- Built-in live code compilation or proctored online exam execution (the portal ingests results exported from external exam platforms or trainer sheets).
- Automated final hiring decisions without human trainer or placement officer review.
- Direct scraping of third-party coding platforms where official exports or APIs are unavailable.
- Payroll, tuition fee management, or general college ERP attendance unrelated to placement training.

### 7.3 Assumptions & Constraints

- Institutional Roll Number is the unique identifier used to match students across different exam and interview spreadsheets; student name alone is never used as a unique key.
- Uploaded spreadsheets may contain arbitrary columns, missing cells, or varied maximum marks, which must be handled gracefully during the upload mapping step.
- If optional AI summarization is used for interview remarks, AI outputs remain suggestions and cannot overwrite official scores without trainer approval.

## 8. Milestone 1 Deliverables Checklist & Mentor Approval

☑ Problem statement (2–3 lines) clearly written and aligned with project goals
☑ Formal SRS document completed for Milestone 1 (Requirement Phase)
☑ User roles identified (Technical Trainer, Student, Admin) with clear permissions
☑ Use case list (10 use cases in Actor -> Goal -> Steps format) and diagram summary included
☑ Functional requirements list ('The system shall...') documented per module
☑ Non-functional requirements list documented across all 6 quality categories
☑ Project scope (In-Scope vs. Out-of-Scope) clearly defined

### Milestone 1 Sign-Off

| Role | Name | Signature | Date |
|---|---|---|---|
| Project Team (Group 171 — 5 Members) | 1. Advay Sharma (2315000160)<br>2. Akanksha Sahu (2315000174)<br>3. Rudra Gupta (2315001894)<br>4. Jatin Kumar Tureha (2315001016)<br>5. Vikram Singh (2315002453) | _____ | _____ |
| Project Mentor / Technical Trainer | Mr. Manwatkar Sumedkumar Janardanji | _____ | _____ |